[redacted]

From: Steven <[redacted]
Sent: Monday, April 01, 2013 11:47 AM
To: [redacted]
Subject: Fwd: pure science...

Sent from my iPhone

Begin forwarded message:

> **From:** "[redacted]
> **Date:** April 1, 2013, 10:00:58 AM EDT
> **To:** Steve Greer <[redacted], [redacted] <[redacted]>, [redacted]
> **Cc:** [redacted]
> **Subject: RE: pure science...**
>
> Steve,
>
> Yes. I am trying to work this out. All I have right now is the "sequence variations" as it's all one can easily move around by internet... about 150MB. The actual alignment is 60 gigabyte and not possible to move wirelessly. I am having someone from my lab run over to the company to pick up the alignment on USB so I can analyze it when I get home tomorrow. I will have more than enough to talk about Wednesday for the camera.
>
> The problem as you relate is the age of the specimen. One can get age & degradation-induced mutation from C-> T and G-> A in old specimens. So any of those mutations we have to be careful about.
>
> More later. Doing my 'day job' today out in Boston.
>
> [redacted]
>
> **From:** Steve Greer [[redacted]
> **Sent:** Sunday, March 31, 2013 10:58 PM
> **To:** [redacted]
> **Cc:** [redacted]
> **Subject:** Re: pure science...
>
> Thanks [redacted] wow.
>
> A few thoughts: even if, of the 1 million mutations, only 1% are real, that equals 10,000 mutations! It is hard to see how a "human", strictly speaking, lives to be 6 (or even 2) years of age with so many mutations.
>
> Did the "DNA clock" indicate an age on the specimen- is it decades old, or centuries- or thousands of years?

Putting on my doctor's/clinician's hat: having delivered and cared for (and resuscitated) premature infants in the ER and NICU, it is hard to fathom how a human, living many years ago in a remote, primitive part of Chile, could survive for a number of years (or even one year) without modern intensive care technologies. To be 6 inches long (equivalent to a 22-24 week fetus) would challenge even our ability today to keep an infant that size alive, never mind decades to centuries ago in a primitive region of the earth. This is not simply a grossly deformed fetus, since bone density indicates years of development (or the earth equivalent...). Any yet...

Another question is: how many mutations can exist in a known species and that species be able to survive past gestation and in the world for a number of years? I know some mutations are lethal, some are not- but 10,000? I do not know the answer to that question.

Again, I guess more questions than answers...
Amazing work- Let's talk soon...Steve

On Mon, Apr 1, 2013 at 12:42 AM, [redacted]
<[redacted] wrote:

Steve,

Well, some first analyses: Looks like we have at least "a" reason (partial) for the dwarfism. Mutations in the gene for PCNT on chromosome 21—known to be associated with osteodysplastic primordial dwarfism.

chr21 19670118 19670118 PRSS7 exonic . 0.89 nonsynonymous
SNV PRSS7:uc002ykw.2:exon19:c.C2194T:p.P732S, rs2824721 Enterokinase deficiency, 226200 (3)|PRSS7, ENTK|606635|21q21.1;

chr21 19770562 19770562 PRSS7 exonic . 0.08 nonsynonymous
SNV PRSS7:uc002ykw.2:exon2:c.A230G:p.K77R, rs2824804 Enterokinase deficiency, 226200 (3)|PRSS7, ENTK|606635|21q21.1;

chr21 27254051 27254051 APP exonic . 0 nonsynonymous
SNV APP:uc011aci.1:exon16:c.A1913C:p.H638P,APP:uc011acg.1:exon8:c.A767C:p.H256P,APP:uc002ymb.2:exon16:c.A2018C:p.H673P,APP:uc002ylz.2:exon18:c.A2243C:p.H748P,APP:uc002yma.2:exon17:c.A2186C:p.H729P,APP:uc011ach.1:exon17:c.A2075C:p.H692P,APP:uc010glj.2:exon15:c.A1850C:p.H617P,APP:uc010glk.2:exon17:c.A2171C:p.H724P, . Alzheimer disease 1, familial, 104300 (3)|APP, AAA, CVAP, AD1|104760|21q21.3;Cerebral amyloid angiopathy, Dutch, Italian, Iowa, Flemish, Arctic variants, 605714 (3)|APP, AAA, CVAP, AD1|104760|21q21.3;

chr21 34614255 34614255 IFNAR2 exonic . 0.87 nonsynonymous
SNV IFNAR2:uc002yrc.2:exon2:c.T28G:p.F10V,IFNAR2:uc002yrd.2:exon2:c.T28G:p.F10V,IFNAR2:uc002yrb.2:exon2:c.T28G:p.F10V,IFNAR2:uc002yrf.2:exon2:c.T28G:p.F10V,IFNAR2:uc002yre.2:exon2:c.T28G:p.F10V, rs1051393 {Hepatitis B virus, susceptibility to}, 610424 (3)|IFNAR2|602376|21q22.11;

chr21 35821821 35821821 KCNE1 exonic . 0.35 nonsynonymous
SNV KCNE1:uc002ytz.2:exon3:c.A112G:p.S38G,KCNE1:uc010gmp.2:exon2:c.A112G:p.S38G,KCNE1:uc010gmq.2:exon3:c.A112G:p.S38G,KCNE1:uc010gmr.2:exon3:c.A112G:p.S38G,KCNE1:uc010gms.2:exon3:c.A112G:p.S38G, rs1805127 Jervell and Lange-Nielsen syndrome 2, 612347 (3)|KCNE1, JLNS, LQT5, JLNS2|176261|21q22.12;Long QT syndrome-5, 613695 (3)|KCNE1, JLNS, LQT5, JLNS2|176261|21q22.12;

chr21 38852992 38852992 DYRK1A exonic . 0 nonsynonymous
SNV DYRK1A:uc002ywj.2:exon4:c.C353G:p.S118C,DYRK1A:uc002ywk.2:exon4:c.C380G:p.S127C,DYRK1A:uc002ywi.2:exon6:c.C380G:p.S127C,DYRK1A:uc002ywh.1:exon4:c.C266G:p.S89C,DYRK1A:uc002ywm.2:exon4:c.C380G:p.S127C,DYRK1A:uc002ywl.2:exon4:c.C380G:p.S127C, . Mental retardation, autosomal dominant 7, 614104 (3)|DYRK1A, MNBH, MNB, MRD7|600855|21q22.13;

chr21 43161357 43161357 RIPK4 exonic . 0.49 nonsynonymous
SNV RIPK4:uc002yzn.1:exon8:c.A1996G:p.M666V, rs3746891 Popliteal pterygium syndrome 2, lethal type, 263650 (3)|RIPK4, NKRD3, DIK, PPS2|605706|21q22.3;

The above list is an intersection of the genes we found when the mutant list from the specimen was intersected with the database of genes KNOWN mutations that cause KNOWN disorders in humans. It does not mean the mutations we see cause such disorders, only that there is a mutation in that gene that someone once had that causes a problem.

There are all kinds of other mutations and I am diving into it, but the list is long—about (literally) one million mutations scattered across the genome. Probably only 1% are real, but even then tracking them down is going to take (literally) weeks to months. I can only scratch the surface. But now that I have this sequence (and 2 more coming in a couple weeks which should give us a ton more "double checking) I can do a lot.

Here is another tiniest sliver of the mutations and genes associated. Now, MANY OF THESE MIGHT BE SEQUENCING ERRORS so I have to check on them first—each one by hand to check. This is just what the first computer scan through gives (just from chromosome 1):

#chr start end gene_name type rmsk SIFT consequence mutation_info dbsnp132

chr1 10109 10109 NONE(dist=NONE),DDX11L1(dist=1765) intergenic Simple_repeat . . . .

chr1 10177 10177 NONE(dist=NONE),DDX11L1(dist=1697) intergenic Simple_repeat . . . .

chr1 10492 10492 NONE(dist=NONE),DDX11L1(dist=1382) intergenic telo . . . rs55998931

chr1 14907 14907 FLJ00075;DDX11L1,DDX11L9 ncRNA_exonic;downstream rs6682375

chr1 14930 14930 FLJ00075;DDX11L1,DDX11L9 ncRNA_exonic;downstream rs6682385

chr1 15211 15211 FLJ00075;DDX11L1,DDX11L9 ncRNA_exonic;downstream rs11586607

chr1 16288 16288 FLJ00075 ncRNA_exonic rs7359244

chr1 16298 16298 FLJ00075 ncRNA_exonic rs2747966

chr1 16378 16378 FLJ00075 ncRNA_exonic rs3950651

chr1 16495 16495 FLJ00075 ncRNA_exonic rs3210724

chr1 16534 16534 FLJ00075 ncRNA_exonic rs15642

chr1 20144 20144 FLJ00075 ncRNA_intronic CR1 . . . rs77331792

chr1 30923 30923 FLJ00075(dist=1553),F379(dist=3689) intergenic Simple_repeat . . . rs806731

chr1 31029 31029 FLJ00075(dist=1659),F379(dist=3583) intergenic ERVL-MaLR . . . rs2870120 chr1 52238 52238 F379(dist=16157),OR4F5(dist=16853) intergenic Low_complexity . . . rs2691277

chr1 54421 54421 F379(dist=18340),OR4F5(dist=14670) intergenic . . . . .

chr1 54844 54844 F379(dist=18763),OR4F5(dist=14247) intergenic Alu . . . .

chr1 55299 55299 F379(dist=19218),OR4F5(dist=13792) intergenic L2 . . . rs10399749

chr1 55545 55545 F379(dist=19464),OR4F5(dist=13546) intergenic L2 . . . rs3020700

chr1 55926 55926 F379(dist=19845),OR4F5(dist=13165) intergenic rs3020698

chr1 57408 57408 F379(dist=21327),OR4F5(dist=11683) intergenic . . . . .

chr1 57417 57417 F379(dist=21336),OR4F5(dist=11674) intergenic . . . . .

chr1 57952 57952 F379(dist=21871),OR4F5(dist=11139) intergenic rs2691334

chr1 58814 58814 F379(dist=22733),OR4F5(dist=10277) intergenic rs114420996

chr1 58866 58866 F379(dist=22785),OR4F5(dist=10225) intergenic rs76646696

chr1 59040 59040 F379(dist=22959),OR4F5(dist=10051) intergenic rs62637815

chr1 59193 59193 F379(dist=23112),OR4F5(dist=9898) intergenic L2 . . . rs77903044

chr1 60332 60332 F379(dist=24251),OR4F5(dist=8759) intergenic rs62637816

chr1 60351 60351 F379(dist=24270),OR4F5(dist=8740) intergenic rs62637817

chr1 60726 60726 F379(dist=24645),OR4F5(dist=8365) intergenic rs2531295

chr1 61442 61442 F379(dist=25361),OR4F5(dist=7649) intergenic rs2531261

chr1 62738 62738 F379(dist=26657),OR4F5(dist=6353) intergenic rs11810446

chr1 63268 63268 F379(dist=27187),OR4F5(dist=5823) intergenic rs28664618

chr1 63643 63643 F379(dist=27562),OR4F5(dist=5448) intergenic . . . . .

chr1 63671 63671 F379(dist=27590),OR4F5(dist=5420) intergen

chr21 43165935 43165935 RIPK4 exonic . 0.38 nonsynonymous
SNV RIPK4:uc002yzn.1:exon6:c.C920T:p.P307L, . Popliteal pterygium syndrome 2, lethal type, 263650 (3)|RIPK4, NKRD3, DIK, PPS2|605706|21q22.3;

chr21 46311889 46311889 ITGB2 exonic . 0.04 nonsynonymous
SNV ITGB2:uc002zgf.3:exon11:c.C1247T:p.T416M,ITGB2:uc002zgg.2:exon11:c.C1247T:p.T416M,ITGB2:uc002zge.2:exon11:c.C1247T:p.T416M,ITGB2:uc010gpw.2:exon10:c.C1076T:p.T359M,ITGB2:uc002zgd.2:exon10:c.C1247T:p.T416M,ITGB2:uc011afl.1:exon11:c.C1013T:p.T338M, rs113084585 Leukocyte adhesion deficiency, 116920 (3)|ITGB2, CD18, LCAMB, LAD|600065|21q22.3;

chr21 46314907 46314907 ITGB2 exonic . 0.04 nonsynonymous
SNV ITGB2:uc002zgf.3:exon9:c.A1062T:p.Q354H,ITGB2:uc002zgg.2:exon9:c.A1062T:p.Q354H,ITGB2:uc002zge.2:exon9:c.A1062T:p.Q354H,ITGB2:uc010gpw.2:exon8:c.A891T:p.Q297H,ITGB2:uc002zgd.2:exon8:c.A1062T:p.Q354H,ITGB2:uc011afl.1:exon9:c.A828T:p.Q276H, rs235330 Leukocyte adhesion deficiency, 116920 (3)|ITGB2, CD18, LCAMB, LAD|600065|21q22.3;

chr21 46910210 46910210 COL18A1 exonic . 0.46 nonsynonymous
SNV COL18A1:uc002zhi.2:exon19:c.G2521A:p.V841I,COL18A1:uc002zhg.2:exon20:c.G1981A:p.V661I,COL18A1:uc011afs.1:exon19:c.G3226A:p.V1076I, rs62000962 Knobloch syndrome, type 1, 267750 (3)|COL18A1, KNO1|120328|21q22.3;

chr21 46911188 46911188 COL18A1 exonic . 0.12 nonsynonymous
SNV COL18A1:uc002zhi.2:exon21:c.C2657G:p.P886R,COL18A1:uc002zhg.2:exon22:c.C2117G:p.P706R,COL18A1:uc011afs.1:exon21:c.C3362G:p.P1121R, rs79980197 Knobloch syndrome, type 1, 267750 (3)|COL18A1, KNO1|120328|21q22.3;

chr21 47545768 47545768 COL6A2 exonic . 0.09 nonsynonymous
SNV COL6A2:uc002zib.1:exon7:c.G257A:p.R86H,COL6A2:uc002zia.1:exon26:c.G2039A:p.R680H,COL6A2:uc002zhz.1:exon26:c.G2039A:p.R680H,COL6A2:uc002zhy.1:exon26:c.G2039A:p.R680H, rs1042917 Bethlem myopathy, 158810 (3)|COL6A2|120240|21q22.3;Myosclerosis, congenital, 255600 (3)|COL6A2|120240|21q22.3;Ullrich congenital muscular dystrophy, 254090 (3)|COL6A2|120240|21q22.3;

chr21 47546124 47546124 COL6A2 exonic . 0.06 nonsynonymous
SNV COL6A2:uc002zib.1:exon7:c.G613A:p.D205N,COL6A2:uc002zia.1:exon26:c.G2395A:p.D799N,COL6A2:uc002zhz.1:exon26:c.G2395A:p.D799N,COL6A2:uc002zhy.1:exon26:c.G2395A:p.D799N, . Bethlem myopathy, 158810 (3)|COL6A2|120240|21q22.3;Myosclerosis, congenital, 255600 (3)|COL6A2|120240|21q22.3;Ullrich congenital muscular dystrophy, 254090 (3)|COL6A2|120240|21q22.3;

chr21 47558473 47558473 FTCD exonic . 0 nonsynonymous SNV;synonymous
SNV FTCD:uc010gqf.2:exon11:c.C1348G:p.P450A,;FTCD:uc002zig.2:exon12:c.C1392G:p.A464A,FTCD:uc002zif.2:exon12:c.C1392G:p.A464A,FTCD:uc002zih.2:exon12:c.C1392G:p.A464A,FTCD:uc010gqg.1:exon12:c.C999G:p.A333A, rs1047179 Glutamate formiminotransferase deficiency, 229100 (3)|FTCD|606806|21q22.3;

chr21 47777063 47777063 PCNT exonic . 1 nonsynonymous
SNV PCNT:uc002zjj.2:exon13:c.G1757A:p.G586E,PCNT:uc002zji.3:exon13:c.G2111A:p.G704E, rs2839223 Microcephalic osteodysplastic primordial dwarfism, type II, 210720 (3)|PCNT, PCNT2, KEN, SCKL4, MOPD2|605925|21q22.3;

chr21 47787002 47787002 PCNT exonic . 0.52 nonsynonymous
SNV PCNT:uc002zjj.2:exon15:c.T2759C:p.V920A,PCNT:uc002zji.3:exon15:c.T3113C:p.V1038A, rs6518289 Microcephalic osteodysplastic primordial dwarfism, type II, 210720 (3)|PCNT, PCNT2, KEN, SCKL4, MOPD2|605925|21q22.3;

chr21 47831758 47831758 PCNT exonic . 0.51 nonsynonymous
SNV PCNT:uc002zjj.2:exon28:c.C5417T:p.A1806V,PCNT:uc002zji.3:exon28:c.C5771T:p.A1924V, . Microcephalic osteodysplastic primordial dwarfism, type II, 210720 (3)|PCNT, PCNT2, KEN, SCKL4, MOPD2|605925|21q22.3;

chr21 47836403 47836403 PCNT exonic . 0.24 nonsynonymous
SNV PCNT:uc002zjj.2:exon30:c.T6217C:p.S2073P,PCNT:uc002zji.3:exon30:c.T6571C:p.S2191P, rs34151633 Microcephalic osteodysplastic primordial dwarfism, type II, 210720 (3)|PCNT, PCNT2, KEN, SCKL4, MOPD2|605925|21q22.3;

chr21 47850484 47850484 PCNT exonic . 0.02 nonsynonymous
SNV PCNT:uc002zjj.2:exon37:c.G7623C:p.Q2541H,PCNT:uc002zji.3:exon37:c.G7977C:p.Q2659H, rs2070426 Microcephalic osteodysplastic primordial dwarfism, type II, 210720 (3)|PCNT, PCNT2, KEN, SCKL4, MOPD2|605925|21q22.3;

chr21 47851753 47851753 PCNT exonic . 1 nonsynonymous
SNV PCNT:uc002zjj.2:exon38:c.A8021G:p.Q2674R,PCNT:uc002zji.3:exon38:c.A8375G:p.Q2792R, rs2073376 Microcephalic osteodysplastic primordial dwarfism, type II, 210720 (3)|PCNT, PCNT2, KEN, SCKL4, MOPD2|605925|21q22.3;

Just because there is a mutation in the gene, doesn't mean it's THE mutation that has been seen to cause the dwarfism. All we have is a mutation in a gene that is associated with dwarfism. So, a hint at this point. Still a ton of analyses left... tip of the iceberg. And any C -> T mutations or G -> A mutations will be suspect due to "ancient DNA" degradation.